

Reflective Lantern - Innovation Report

New Project Built

Name: Care-Compass

Date: 2026-08-03

Inspired by: MLOps demand for production-grade clinical decision support and readmission-penalty reduction programmes

What it does: Clinical ML API predicting 30-day hospital readmission risk from patient vitals, diagnoses, lab trends and admission history using an XGBoost pipeline with KS-test and PSI drift monitoring, automated retraining, and PostgreSQL-backed prediction logging.

ACTION REQUIRED - Not Published to GitHub

GitHub repository creation was refused by the session gateway: this session is bound to the reflective-lantern repository, so POST /user/repos returns 403 regardless of the PAT. The complete project (all 66 commits plus the v1.0.0 tag) is preserved as a verified git bundle and patch series under pending/ in the reflective-lantern repository.

To publish, run locally:

```
gh repo create Care-Compass --public \
  --description "Clinical ML API predicting 30-day hospital readmission risk..."
git clone pending/Care-Compass-2026-08-03.bundle Care-Compass
cd Care-Compass
git remote set-url origin https://github.com/atharvadevne123/Care-Compass
git push -u origin main --tags
```

Tech Stack Coverage: 100%

- [X] Python 3.11
- [X] FastAPI (6 versioned endpoints)
- [X] XGBoost ensemble
- [X] sklearn Pipeline (26 features)
- [X] Model monitoring (KS + PSI drift)
- [X] Docker + compose + PostgreSQL
- [X] SQLAlchemy (SQLite dev / Postgres prod)
- [X] pytest suite (75 tests)
- [X] .env.example
- [X] GitHub Actions CI (ruff + pytest)

Optional Enhancements (7 of 9 categories)

- [X] Automated retraining pipeline (AUC-regression guarded)
- [X] Drift detection (KS-test + Population Stability Index)
- [X] 5-fold stratified CV with AUC-ROC
- [X] Alembic database migrations (verified up and down)
- [X] Time-series lag and rolling-vitals features
- [X] Structured JSON logging with correlation IDs
- [X] Per-client rate limiting

Project Stats

Files created: 39

Tests written: 75 (all passing)

Commits: 66

Cross-validated AUC-ROC: 0.777 (5-fold stratified)

ruff check: clean

Verification Performed

- Server booted; all 6 endpoints returned live responses
- High-utilisation patient scored 0.987, low-risk patient 0.020
- Feature importances clinically sensible (medications, prior admissions, SNF discharge)
- Alembic migration verified upgrade to head and downgrade to base
- Retraining pipeline executed end to end
- Two defects found in live testing and fixed (correlation ID loss, PSI small-sample)